

Stat 17 Week 6

Continuous Distributions

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Continuous Distributions & Standard Normal

Empirical Rule (68–95–99.7 Rule)

- Applies to bell-shaped (approximately normal) distributions
- About 68% of data fall within $\mu \pm 1\sigma$
- About 95% of data fall within $\mu \pm 2\sigma$
- About 99.7% of data fall within $\mu \pm 3\sigma$
- Beyond 3 standard deviations:
 - Total outside area $\approx 0.3\%$
 - Each tail $\approx 0.15\%$

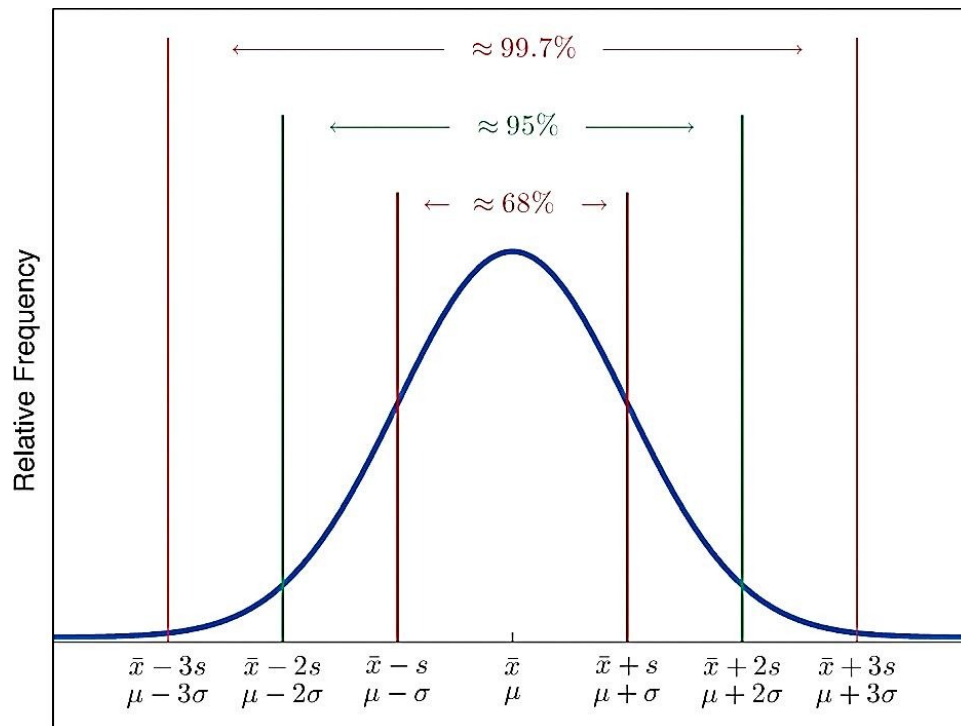


Figure 1: Visualization of the Empirical Rule

Uniform Distribution

- If $X \sim U(a, b)$, then probability is based on interval length

$$P(c < X < d) = \frac{d - c}{b - a}$$

- All values in (a, b) are equally likely
- Graph is a rectangle (constant density)

Normal Distribution

- Bell-shaped and symmetric
- Completely determined by:
 - Mean μ
 - Standard deviation σ
- Standard normal distribution:

$$Z \sim N(0, 1)$$

Z-Score

- Standardization formula:

$$z = \frac{x - \mu}{\sigma}$$

- Measures how many standard deviations a value is from the mean
- Used to compare relative performance
- More unusual values have larger $|z|$

Converting to Standard Normal

- If $X \sim N(\mu, \sigma)$, convert using:

$$Z = \frac{X - \mu}{\sigma}$$

- After converting, use z-table

Finding Probabilities

- Left tail:

$$P(X < a)$$

- Right tail:

$$P(X > a) = 1 - P(X < a)$$

- Between two values:

$$P(a < X < b) = P(z_1 < Z < z_2)$$

Percentiles

- To find a percentile:
 1. Find corresponding z-score from table
 2. Convert back:

$$x = \mu + z\sigma$$

Common Z-Values

- 95% confidence level: $z = 1.96$
- Middle 99%: $z = \pm 2.575$
- 90th percentile: $z = 1.28$
- 75th percentile: $z = 0.67$

Two Types of Z-Tables

- Table A: $T(z) = P(0 < Z < z)$
- Table B: $\Phi(z) = P(Z < z)$
- Relationship (for $z > 0$):

$$\Phi(z) = 0.5 + T(z)$$